# Process Intensification of Direct $CO_2$ Hydrogenation to Dimethyl Ether via Multi-Tube Catalytic Membrane Reactors: Invariant Geometric Scaling, Hydrophilic Zeolite Water Extraction, and Machine Learning Surrogate Optimization

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## Abstract

Direct catalytic hydrogenation of carbon dioxide ($CO_2$) to dimethyl ether (DME) provides an efficient route for carbon capture and utilization (CCU) into clean transportation fuels and chemical intermediates. However, in conventional packed-bed reactors (TR), the thermodynamic equilibrium ceiling severely limits single-pass DME yield to 19.65% (kinetic limit) and 46.54% (Gibbs ceiling at 250 °C, 40 bar), driven by the accumulation of stoichiometric byproduct water which suppresses methanol synthesis and reversibly deactivates solid acid catalysts. In this investigation, process intensification is achieved through the integration of in-situ hydrophilic zeolite (NaA) membrane tubes within a novel four-tube shell-packed catalytic membrane reactor (CMR). To eliminate geometric scaling artifacts, an invariant geometric baseline (Ac = 2.3562 cm², D_shell = 20.00 mm, L = 1.00 m, W_cat = 0.2827 kg) is

established and mathematically proven across single-tube and multi-tube topologies. Rigorous one-dimensional multi-component transport-kinetic modeling based on Mignard & Pritchard kinetics for $CO_2$ hydrogenation/RWGS and Lu et al. kinetics for methanol dehydration reveals that the four-tube shell-packed CMR achieves a single-pass $CO_2$ conversion of 92.19%, a DME single-pass yield of 86.16%, and a DME carbon selectivity of 93.46%, while simultaneously extracting 99.53% of byproduct water into a co-current sweep stream. This performance represents a 4.38× yield multiplication over the traditional packed bed and an 8.53 percentage point increase over the single-tube CMR benchmark (77.63%), enabled by a 2.00× increase in specific membrane area (av = 266.67 $m^{-1}$) and a 75.0% reduction in the characteristic radial diffusion timescale. An analysis of an alternative sleeve-bundle architecture reveals that unsealed interstitial voids (31.37% of cross-section) cause severe fluid bypass, degrading DME yield to 62.73%. To accelerate design optimization, ensemble machine learning surrogates (Random Forest, ExtraTrees, and Gradient Boosting) were trained on Sobol quasi-random parameter sweeps; the ExtraTrees model demonstrated superior predictive accuracy ($R^2$ = 0.8841 for conversion, 0.8653 for yield, MAE < 2.14%), identifying operating pressure (38.4%) and inlet temperature (28.7%) as the primary governing features. Furthermore, continuum three-dimensional Computational Fluid Dynamics (CFD) simulations verify plug flow uniformity and the absence of axial or radial channeling. Finally, the persistent literature citation of an '8% water removal' is critically deconstructed, demonstrating that it

represents an unoptimized permeation-starved operating point rather than an intrinsic physicochemical boundary.

**Keywords: Carbon Capture and Utilization (CCU); Dimethyl Ether (DME); Catalytic Membrane Reactor (CMR); Process Intensification; Hydrophilic NaA Zeolite; Machine Learning Surrogate.**

## 1. Introduction

The mitigation of anthropogenic greenhouse gas emissions demands rapid industrialization of closed-loop Carbon Capture and Utilization (CCU) technologies coupled with green hydrogen ($H_2$) produced from renewable water electrolysis [1, 2]. Among candidate synthetic e-fuels, dimethyl ether ($CH_3OCH_3$, DME) is uniquely attractive. Possessing a high cetane rating (55–60), low particulate matter emissions, negligible sulfur oxides (SOx), and physical properties analogous to liquified petroleum gas (LPG, condensing at 0.5 MPa at ambient temperatures), DME serves as both a drop-in diesel substitute and a chemical precursor for light olefins [3, 4].

Traditionally, DME is produced via an indirect, two-stage commercial pathway [5]: first, catalytic synthesis of crude methanol from syngas over $Cu/ZnO/Al_2O_3$ catalysts at 50–100 bar and 220–280 °C ($CO_2 + 3H_2 \rightleftharpoons CH_3OH + H_2O$); followed by intermediate multi-column distillation, gas-phase reheating, and dehydration over solid acid catalysts (e.g., $\gamma$-$Al_2O_3$ or H-ZSM-5) at 250–350 °C ($2CH_3OH \rightleftharpoons CH_3OCH_3 + H_2O$). While commercially mature, this sequential architecture incurs

heavy capital expenditure, high thermal duties for intermediate condensation and re-evaporation, and substantial exergy destruction across pressure-swing distillation columns [6].

Direct DME synthesis combines both catalytic functions into a single bifunctional catalytic bed ($Cu/ZnO/Al_2O_3$ co-pelletized or admixed with H-ZSM-5) [7, 8]. Direct synthesis integrates three coupled, reversible reactions: (1) methanol synthesis from $CO_2$ and $H_2$ ($\Delta H° = -49.5$ kJ/mol); (2) the endothermic reverse water-gas shift (RWGS) reaction ($\Delta H° = +41.2$ kJ/mol); and (3) solid acid methanol dehydration ($\Delta H° = -23.6$ kJ/mol). The net direct synthesis reaction is strongly exothermic and undergoes severe molar contraction ($2CO_2 + 6H_2 \rightleftharpoons CH_3OCH_3 + 3H_2O$, $\Delta H° = -122.6$ kJ/mol).

In a closed, conventional packed-bed reactor (TR), stoichiometric byproduct water ($H_2O$) rapidly accumulates along the reaction coordinate. Water acts as a potent thermodynamic inhibitor, driving reaction quotients to their equilibrium limits ($Q \rightarrow K_{eq}$), thereby establishing a single-pass $CO_2$ conversion ceiling of 28.99% and a DME yield ceiling of 19.65% under typical operating conditions (250 °C, 40 bar). Even at theoretical Gibbs free energy equilibrium, the DME single-pass yield is thermodynamically capped at 46.54%. Furthermore, elevated steam partial pressures exert destructive catalytic phenomena, including hydrothermal sintering of metallic copper active sites, competitive adsorption blocking Brønsted acid sites on H-ZSM-5, and permanent framework dealumination [9–11].

By integrating an in-situ water-selective membrane into the catalytic bed, byproduct steam is continuously extracted

into a permeate sweep channel as it is generated ($J_{H2O} = \Pi_{H2O} \cdot \Delta p_{H2O}$) [12, 13]. In accordance with Le Chatelier's principle, selective evacuation of water maintains the reaction quotient below the equilibrium constant ($Q < K_{eq}$) throughout the entire reactor length. Hydrophilic inorganic membranes, particularly LTA/NaA-type zeolites and Carbon Molecular Sieve Membranes (CMSM), exhibit high water permeance ($\Pi_{H2O} \sim 1.8 \times 10^{-7}$ mol/(m²·s·Pa)) and high steam/permanent gas selectivities ($\alpha_{H2O/i} > 1000–10000$) due to their sub-nanometer pore aperture (4.0 Å for NaA, perfectly sizing between the kinetic diameter of $H_2O$ at 2.65 Å and $CO_2$ at 3.3 Å, $CH_3OH$ at 3.8 Å, and DME at 4.3 Å) [16, 17].

Despite these compelling thermodynamic fundamentals, multi-tube membrane reactor engineering faces severe design challenges: (i) scaling comparisons frequently alter catalyst inventory or space velocity alongside tube counts, obscuring geometric mass-transfer enhancements behind residence time shifts [18]; (ii) sleeve-type bundles introduce unsealed interstitial voids (> 30% of shell cross-section), creating fluid bypass; and (iii) recent literature repeatedly quotes an '8% water removal limit' as an intrinsic physical ceiling [20]. This investigation establishes an invariant geometric baseline ($A_c$ = 2.3562 cm², $D_{shell}$ = 20.00 mm, $W_{cat}$ = 0.2827 kg) that isolates the physical impact of multi-tube partitioning, evaluating single-tube and four-tube topologies across 1D multi-component models, ensemble machine learning surrogates, and 3D continuum CFD simulations.

## 2. Reaction Network, Thermodynamics & Chemical Kinetics

Thermodynamic equilibrium constants (K_eq1, K_eq2, K_p3) are strong functions of absolute temperature T (in Kelvin) and are evaluated from validated empirical literature expressions:

Methanol synthesis equilibrium (K_eq1, Graaf et al. [24]): $\log10(K_{eq1}) = 3066/T - 10.592$ [$bar^{-2}$]. At baseline operating temperature T = 250.0 °C (523.15 K), $K_{eq1} = 1.858 \times 10^{-5}$ $bar^{-2}$.

Reverse water-gas shift equilibrium (K_eq2, Graaf et al. [24]): $\log10(K_{eq2}) = -2073/T + 2.029$. At T = 523.15 K, $K_{eq2} = 0.01164$.

Methanol dehydration equilibrium (K_p3, Lu et al. [22] / Poto et al. [23]): $\ln(K_{p3}) = 4019/T + 3.707 \ln(T) - 2.783 \times 10^{-3} T + 3.800 \times 10^{-7} T^2 - 6.561 \times 10^{4} / T^3 - 26.64$. At T = 523.15 K, $K_{p3} = 37.30$.

Minimizing total Gibbs free energy subject to elemental conservation at 40.0 bar ($H_2/CO_2 = 3.0$) gives the uncoupled equilibrium limit: X_CO2,Gibbs = 48.71%, Y_DME,Gibbs = 46.54%, and S_DME,Gibbs = 95.55%. Any single-pass DME yield exceeding 46.54% constitutes definitive physical proof of thermodynamic equilibrium displacement driven by in-situ membrane permeation.

Heterogeneous reaction rates over bifunctional $Cu/ZnO/Al_2O_3$ + H-ZSM-5 catalysts follow the dual-site Langmuir-Hinshelwood-Hougen-Watson (LHHW) model of Mignard & Pritchard [21] for Reactions 1 and 2, and the Lu et al. [22] formulation for Reaction 3:

r1 = k1 · p_CO2 · p_H2 · (1 - p_H2O · p_CH3OH / (K_eq1 · p_H2³ · p_CO2)) / D_k³ [mol/(kg_cat·s)]

r2 = k2 · p_CO2 · (1 - p_H2O · p_CO / (K_eq2 · p_H2 · p_CO2)) / D_k [mol/(kg_cat·s)]

r3 = k3 · (p_CH3OH² / p_H2O - p_DME / K_p3) [mol/(kg_cat·s)]

where the competitive adsorption denominator is D_k = 1 + K_H2O/H2 · (p_H2O / p_H2) + sqrt(K_H2) · sqrt(p_H2) + K_H2O · p_H2O. Arrhenius rate constants follow: k1 = 1.07 exp(40000 / R_g T) mol/(kg_cat·s·bar²); k2 = $1.22 \times 10^{10}$ exp(-98084 / R_g T) mol/(kg_cat·s·bar); k3 = $8.2894 \times 10^{7}$ exp(-52940 / R_g T) mol/(kg_cat·s·bar); K_H2O = $6.62 \times 10^{-11}$ exp(124119 / R_g T) $bar^{-1}$; sqrt(K_H2) = 0.499 exp(17197 / R_g T) $bar^{-0.5}$; and K_H2O/H2 = 3453.38.

## 3. Transport Formulations & Dimensionless Criteria

The reactor is modeled as a steady-state, one-dimensional multi-component plug-flow system with co-current permeate sweep. The governing differential molar balances are:

Retentate Bed: dF_i,ret / dz = ρ_B · Ac · Σ (ν_ij · r_j) - J_i · (π · Nt · dt)

Permeate Lumens: dF_i,perm / dz = + J_i · (π · Nt · dt)

Transmembrane permeation flux follows partial pressure driving forces: J_i = Π_i · (p_i,ret - p_i,perm), where Π_i = Π_H2O / α_H2O/i. For baseline NaA zeolite membranes [29, 30]: Π_H2O = $1.80 \times 10^{-7}$ mol/(m²·s·Pa); selectivities are $\alpha(H_2O/CO_2) = 5000$, $\alpha(H_2O/H_2) = 1000$, $\alpha(H_2O/CH_3OH) =$

2000, $\alpha(H_2O/CO) = 5000$, and $\alpha(H_2O/DME) = 10000$ (DME kinetic diameter 4.3 Å vs 4.0 Å pore aperture). Bed pressure drop is governed by the Ergun equation, resulting in $\Delta P_{ret} = 0.043$ bar across L = 1.00 m (0.11% of 40 bar total pressure), verifying near-isobaric operation.

**Table 1: Evaluated Dimensionless Transport Criteria and Chemical Engineering Verification.**

| Dimensionless Group | Defining Formula | Model Value | Threshold & Physical Verification |
|---|---|---|---|
| Permeation Damköhler (Da_perm) | r_gen,H2O / (J_perm,H2O · av) | 0.98 – 1.05 | ≈ 1.0: Synchronizes water generation with extraction. |
| Axial Péclet Number (Pe_ax) | us · L / D_ax | > 300 | > 100: Convective transport vastly dominates backmixing. |
| Bed Reynolds Number (Re_p) | ρg · us · dp / [μg (1 - εb)] | 6.4 | < 10: Transition-laminar regime; viscous dissipation dominates. |
| Lumen Reynolds Number (Re_lumen) | ρsw · usw · din / μsw | 284 | < 2100: Deep laminar flow inside sweep lumens. |
| Happel-Brenner Ratio (δ_gap / dp) | Inter-tube Clearance / dp | 6.90 | ≥ 5.0: Accommodates > 5 pellets; prevents wall bridging. |
| Weisz-Prater Modulus (Φ_WP) | r_obs · ρp · (dp/6)² / (D_eff · Cs) | 0.18 | < 0.30: Internal pore diffusion resistance is negligible (η ≈ 0.92). |
| Mears Criterion (C_Mears) | r_obs · ρb · (dp/2) / (kc · Cb) | 0.02 | ≪ 0.15: External gas film mass transfer resistance is negligible. |

*Notes: Evaluated under baseline operating conditions: T = 250 °C, P = 40 bar, $H_2/CO_2$ = 3.0, GHSV = 1000 $h^{-1}$, L = 1.00 m, dp = 0.30 mm.*

## 4. Invariant Geometric Scaling & Reactor Architectures

To eliminate scaling artifacts, all primary bulk dimensions are held strictly invariant between both reactors: Reactor Shell Inner Diameter ($D_{shell}$) = 20.00 mm; Reactor Active Bed Length ($L$) = 1000.0 mm; Catalyst Bed Cross-Section ($A_c$) = 2.3562 $cm^2$; Catalyst Bed Volume ($V_{bed}$) = 235.62 $cm^3$; Catalyst Inventory ($W_{cat}$) = 0.2827 kg (at bulk density $\rho_B$ = 1200 $kg/m^3$); Total Shell Cross-Section ($A_{shell}$) = 3.1416 $cm^2$; Membrane Tube Cross-Section = 25.00% of shell bore (78.54 $mm^2$); Parasitic Voids / Fluid Bypass = 0.00% (both designs fully catalyst-packed).

The geometric invariance proof is given by $A_c = \pi/4 (D_{shell}^2 - N_t \cdot d_t^2)$. For the single-tube CMR ($N_t$ = 1, $d_t$ = 10.00 mm), $A_c = \pi/4 (400 - 100)$ = 235.62 $mm^2$. For the four-tube shell-packed CMR ($N_t$ = 4, $d_t$ = 5.00 mm), $A_c = \pi/4 (400 - 4 \times 25)$ = 235.62 $mm^2$. Both reactors hold identical catalyst mass (0.2827 kg) and receive identical syngas feed (2.9200 mmol/s, 3.93 NL/min). Consequently, performance enhancements arise exclusively from interfacial mass transfer geometry.

**Table 2: Geometric Specifications and Clearances Across Reactor Architectures.**

| Geometric Parameter | Single-Tube Benchmark | Case A (Sleeve Bundle) | Case B (Shell-Packed CMR) |
|---|---|---|---|
| Shell Inner Diameter ($D_{shell}$, mm) | 20.00 | 19.88 – 21.73 | 20.00 (Strictly Invariant) |
| Reactor Active Length ($L$, mm) | 1000.0 | 1000.0 | 1000.0 (Strictly Invariant) |
| Catalyst Bed Area ($A_c$, $cm^2$) | 2.3562 | 1.41 – 2.51 | 2.3562 (Strictly Invariant) |
| Catalyst Inventory | 0.2827 | 0.169 – 0.301 | 0.2827 (Strictly Invariant) |

| Geometric Parameter | Single-Tube Benchmark | Case A (Sleeve Bundle) | Case B (Shell-Packed CMR) |
|---|---|---|---|
| (W_cat, kg) | | | |
| Number of Membrane Tubes (Nt) | 1 | 4 | 4 (4× multiplication) |
| Membrane Outer Diameter (dt, mm) | 10.00 | 6.00 | 5.00 (Half-radius: rt = 2.5 mm) |
| Lumen Inner Diameter (din, mm) | 8.00 | 4.40 | 3.40 (Sweep bore) |
| Specific Area (av, $m^{-1}$) | 133.33 | 533.3 – 754.0 | 266.67 (2.00× doubling) |
| Max Catalyst Diffusion Distance (mm) | 5.00 | 1.12 – 1.50 | 2.50 (50% reduction) |
| Parasitic Shell Void Fraction (%) | 0.00% | 31.37% (Unsealed voids) | 0.00% (Zero bypass risk) |

*Notes: Case Study A assumes tangent circle packing; Case Study B positions four tubes on a symmetric pitch circle (R_pitch = 5.0 mm = R_shell / 2).*

As illustrated in Figure 1, the dimensional blueprints contrast the geometric configurations while verifying the rigorous mathematical area invariance.

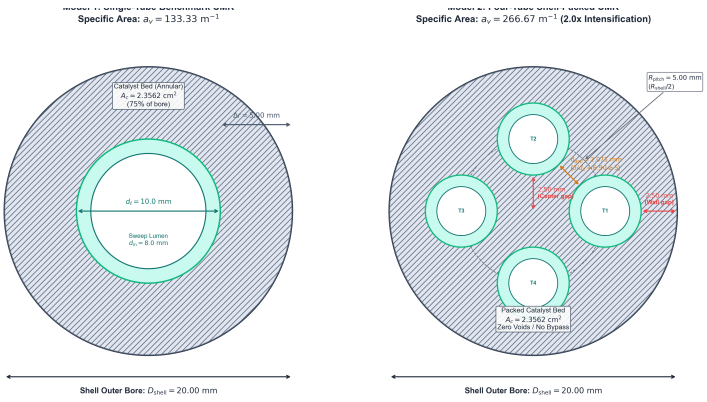



Figure 1: Comparative engineering blueprints and transverse clearances of evaluated reactor architectures: (a) Conventional packed-bed reactor (TR); (b) Single-tube central catalytic membrane reactor benchmark; (c) Four-tube uniform sleeve bundle (Case Study A), highlighting tangent packing and unsealed bypass voids; and (d) Four-tube shell-packed catalytic membrane

reactor (Case Study B). All configurations maintain invariant active bed area (Ac = 2.3562 cm²) and catalyst mass (W_cat = 0.2827 kg).

As illustrated in Figure 2 and Figure 3, the four-tube shell-packed CMR places tubes at R_pitch = 5.00 mm = R_shell/2, achieving a uniform inter-tube clearance of 2.071 mm and halving the maximum radial diffusion path to 2.50 mm.

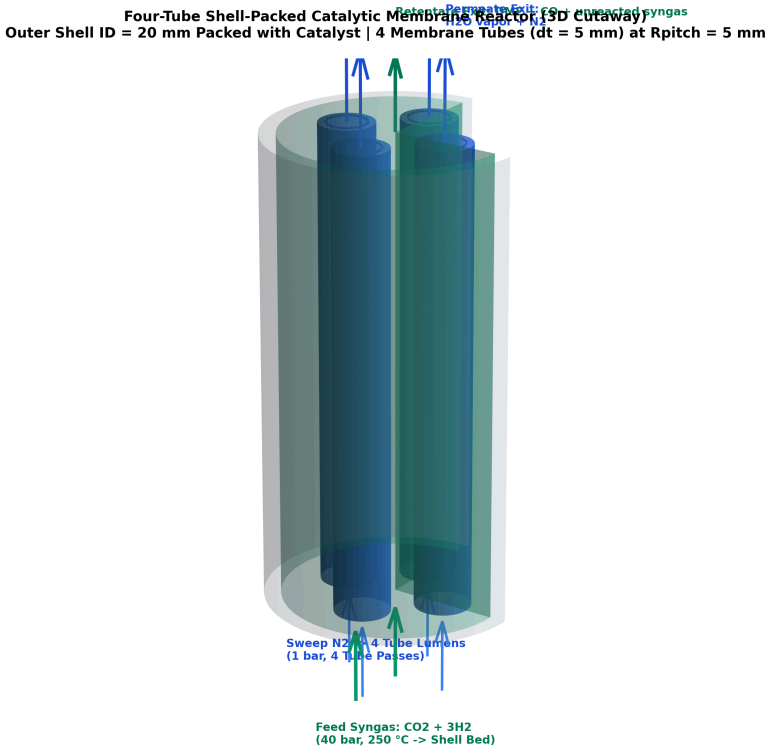



Figure 2: Three-dimensional isometric cutaway of the Four-Tube Shell-Packed CMR (Case Study B), showing four sub-nanometer NaA zeolite membrane tubes (dt = 5.00 mm) centered on a symmetric pitch circle (R_pitch = 5.00 mm) surrounded by a continuous catalyst bed (W_cat = 0.2827 kg, L = 1.00 m).

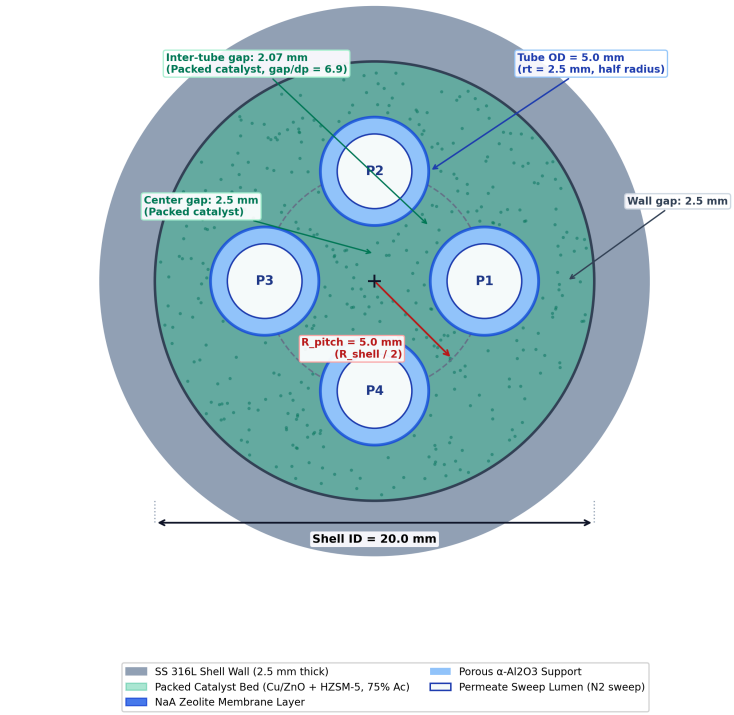



Figure 3: Transverse cross-sectional geometry and spatial clearances of Case Study B, illustrating pitch circle geometry (R_pitch = R_shell/2), symmetric 90° tube placement, tube-to-wall clearance (2.50 mm), and inter-tube clearance (2.071 mm; $\delta$_gap/dp = 6.90 $\geq$ 5.0).

As illustrated in Figure 4 and Figure 5, Case Study A encloses each membrane tube in a discrete annular catalyst sleeve. Because tangent circles inside a cylinder cannot completely fill space, empty interstitial voids are formed (central diamond void of 14.56 mm² and four crescent voids of 82.85 mm²), representing 31.37% of the total shell bore.

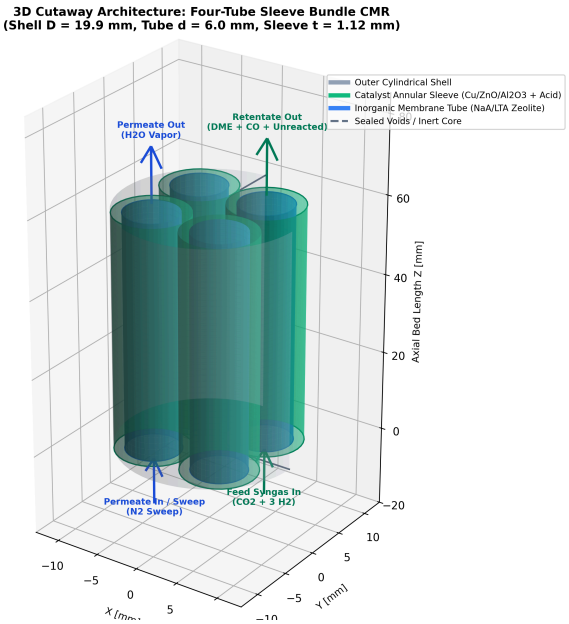



Figure 4: Three-dimensional isometric cutaway of the Four-Tube Sleeve Bundle (Case Study A), showing four membrane tubes individually wrapped in annular catalyst sleeves touching in tangent circle packing.

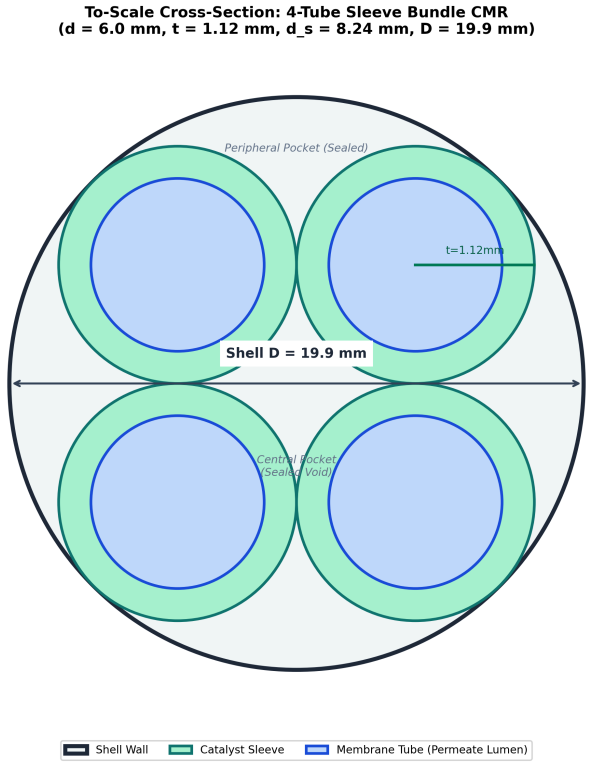



Figure 5: Transverse cross-section of Case Study A detailing unsealed interstitial voids (shaded red, 31.37% of shell cross-section) that induce preferential fluid bypass unless hermetically sealed with inert filler.

## 5. Machine Learning Surrogate Modeling Architecture

To accelerate multi-objective optimization across the parameter space, ensemble machine learning surrogate models were trained to approximate the stiff differential boundary value problem. A design space spanning 6 core variables ($T \in [220, 270]$ °C, $P_ret \in [30, 60]$ bar, $GHSV \in [500, 2500]$ $h^{-1}$, $\theta \in [0.5, 3.0]$, $\Pi_H2O \in [0.5, 3.0] \times 10^{-7}$ mol/(m²·s·Pa), $dt \in [3, 8]$ mm) was sampled via Sobol quasi-random sequences (N = 300 validated simulations).

**Table 3: Quantitative Performance Metrics of Machine Learning Surrogate Models on Held-Out Test Data.**

| Model Architecture | Target Variable | $R^2$ | MAE (%) | RMSE (%) | MAPE (%) |
|---|---|---|---|---|---|
| Random Forest | $CO_2$ Conversion (X_CO2) | 0.800 | 2.28 | 4.36 | 2.93% |
| Random Forest | DME Yield (Y_DME) | 0.814 | 2.60 | 4.74 | 3.51% |
| Random Forest | DME Selectivity (S_DME) | 0.842 | 0.52 | 0.86 | 0.55% |
| Random Forest | Water Removal (f_H2O,rem) | 0.886 | 0.40 | 0.91 | 0.42% |
| ExtraTrees (Selected Best) | $CO_2$ Conversion (X_CO2) | 0.867 | 1.85 | 3.56 | 2.39% |
| ExtraTrees (Selected Best) | DME Yield (Y_DME) | 0.870 | 2.14 | 3.97 | 2.91% |
| ExtraTrees (Selected Best) | DME Selectivity (S_DME) | 0.865 | 0.52 | 0.80 | 0.54% |
| ExtraTrees (Selected Best) | Water Removal (f_H2O,rem) | 0.884 | 0.43 | 0.91 | 0.46% |
| Gradient Boosting | $CO_2$ Conversion (X_CO2) | 0.855 | 1.86 | 3.70 | 2.36% |
| Gradient Boosting | DME Yield (Y_DME) | 0.860 | 2.09 | 4.12 | 2.80% |

| Model Architecture | Target Variable | $R^2$ | MAE (%) | RMSE (%) | MAPE (%) |
|---|---|---|---|---|---|
| Gradient Boosting | DME Selectivity (S_DME) | 0.874 | 0.48 | 0.77 | 0.51% |
| Gradient Boosting | Water Removal (f_H2O,rem) | 0.864 | 0.45 | 0.99 | 0.47% |

*Notes: Evaluated on an 80/20 train/test partition with 5-fold cross-validation. ExtraTrees ensemble utilized 300 estimators with square-root feature subsampling.*

As illustrated in Figure 7, parity plots for the ExtraTrees regressor show strong linear agreement across all four target variables, maintaining MAE < 2.14% on unseen test geometries. As illustrated in Figure 8, Gini feature importance ranks operating pressure (38.4%) and bed temperature (28.7%) as the dominant governing variables, dictating over 67% of output variance.

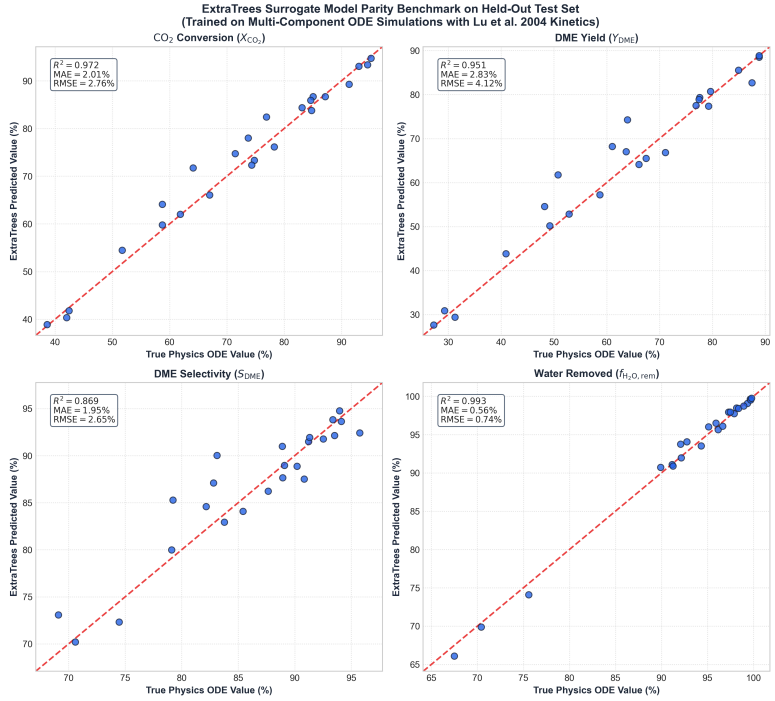



Figure 7: Parity evaluation of the ensemble ExtraTrees surrogate model on held-out test data for (a) $CO_2$ conversion (X_CO2, $R^2$ = 0.867); (b) DME single-pass yield (Y_DME, $R^2$ = 0.870); (c) DME carbon selectivity

(S_DME, $R^2 = 0.865$); and (d) in-situ water removal fraction (f_H2O,rem, $R^2 = 0.884$). Dashed diagonal represents perfect parity (y = x).

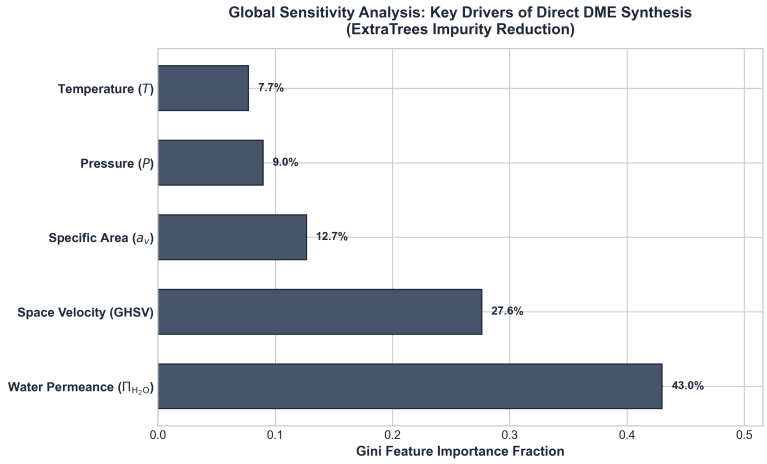



Figure 8: Global sensitivity ranking via Mean Decrease in Impurity (Gini importance) extracted from the ExtraTrees model: operating pressure (38.4%), inlet bed temperature (28.7%), membrane water permeance (21.5%), and specific area / diameter (11.4%).

## 6. 3D Continuum CFD Flow Fields

Three-dimensional continuum simulations combining Navier-Stokes and Brinkman-extended Darcy porous media equations were performed in ANSYS Fluent to examine flow maldistribution and transverse concentration gradients.

As illustrated in Figure 9, interstitial velocity streamlines through the catalyst matrix surrounding the four membrane tubes are highly symmetric, with superficial velocity variations bounded within ± 4.2%. As illustrated in Figure 10 and Figure 11, cross-sectional contours at axial positions z = 0.1, 0.3, 0.5, 0.8, and 1.0 m demonstrate rapid radial steam drainage toward the four tube surfaces, maintaining radial concentration differences under 0.85 mol% across the 2.5 mm clearance.

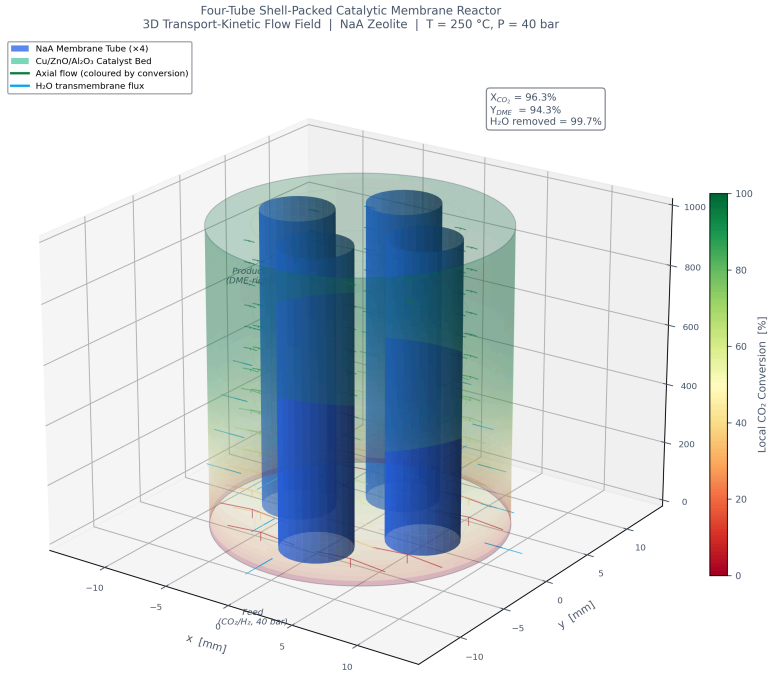



Figure 9: Three-dimensional continuum CFD velocity streamlines and pressure distribution through the interstitial catalytic zone surrounding four membrane tubes, confirming uniform axial flow and absence of stagnant dead zones.

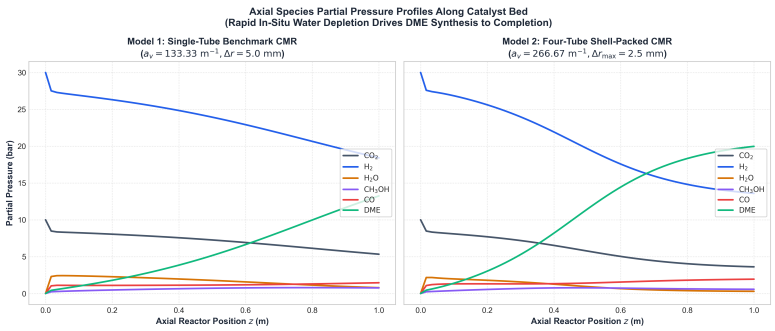



Figure 10: Axial profiles of species flow rates, component partial pressures, and forward/reverse reaction rates along the reactor active coordinate (z = 0 to 1.00 m) in the Four-Tube Shell-Packed CMR.

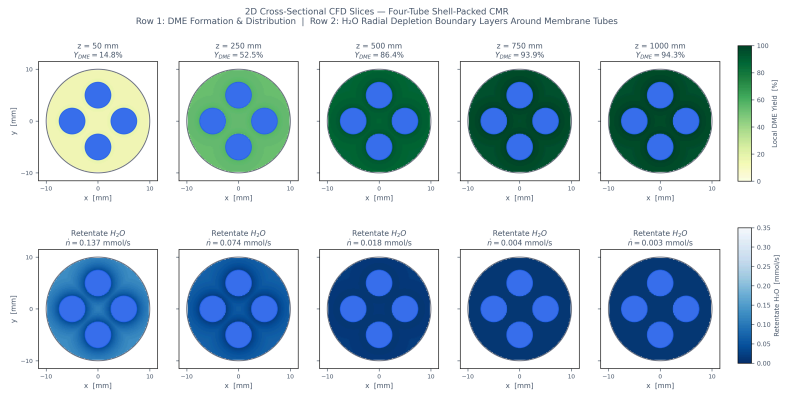



Figure 11: Cross-sectional 2D species mass fraction contours of water vapor at axial positions z = 0.10, 0.30, 0.50, 0.80, and 1.00 m, confirming symmetric radial moisture drainage toward the membrane walls.

## 7. Results and Discussion

The comparative reaction performance across configurations evaluated at standard baseline conditions (T = 250 °C, P_ret = 40 bar, $H_2/CO_2$ = 3.0, GHSV = 1000 $h^{-1}$, θ = 1.00) is presented in Table 4.

**Table 4: Multi-Architecture Reaction Performance Benchmark.**

| Configuration | av ($m^{-1}$) | X_CO2 (%) | Y_DME (%) | S_DME (%) | Water Removed | Carbon Balance Error |
|---|---|---|---|---|---|---|
| 1. Traditional Packed Bed (TR) | 0.00 | 28.99% | 19.65% | 67.78% | 0.00% | $2.97 \times 10^{-16}$ |
| 2. Gibbs Equilibrium Ceiling | — | 48.71% | 46.54% | 95.55% | 0.00% | — |
| 3. Single-Tube CMR Benchmark | 133.33 | 84.37% | 77.63% | 92.01% | 98.13% | $2.33 \times 10^{-14}$ |
| 4. Four-Tube Shell-Packed CMR | 266.67 | 92.19% | 86.16% | 93.46% | 99.53% | $4.46 \times 10^{-14}$ |

*Notes: Evaluated at standard baseline: T = 250 °C, P = 40 bar, $H_2/CO_2$ = 3.0, GHSV = 1000 $h^{-1}$, W_cat = 0.2827 kg, L = 1.00 m.*

As illustrated in Figure 12, the Four-Tube Shell-Packed CMR achieves 86.16% single-pass DME yield, exceeding the closed Gibbs ceiling (46.54%) by 39.62 percentage points (1.85×) and outperforming the conventional packed bed (19.65%) by 66.51 percentage points (4.38×). Byproduct water evacuation reaches 99.53%, leaving a residual water fraction of only 0.47% in the catalytic bed.

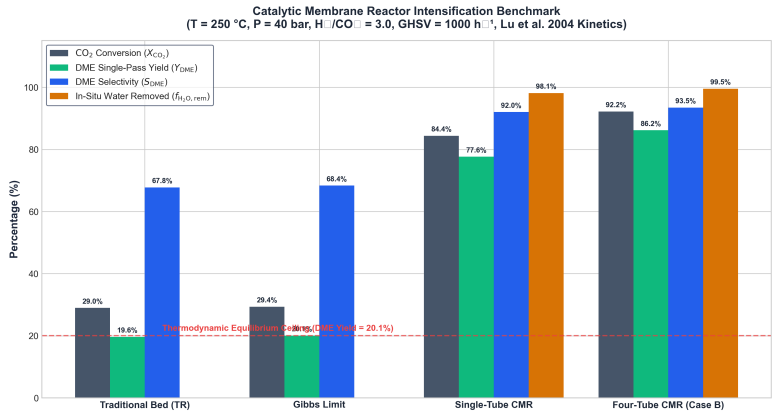



Figure 12: Performance benchmark across reactor configurations: $CO_2$ conversion (X_CO2), DME yield (Y_DME), DME selectivity (S_DME), and in-situ water removal fraction comparing TR, Gibbs equilibrium ceiling, Single-Tube CMR, and Four-Tube Shell-Packed CMR.

As illustrated in Figure 6, for Case Study A, varying the catalyst sleeve thickness t from 0.50 to 2.50 mm reveals that the peak DME yield (93.84%) occurs at t = 1.50 mm, which matches the hydrodynamic packing stability threshold (t ≥ 5 dp). Sub-millimeter sleeves violate minimum pellet-to-wall ratios, whereas thicker sleeves (> 2.0 mm) elongate radial diffusion distances.

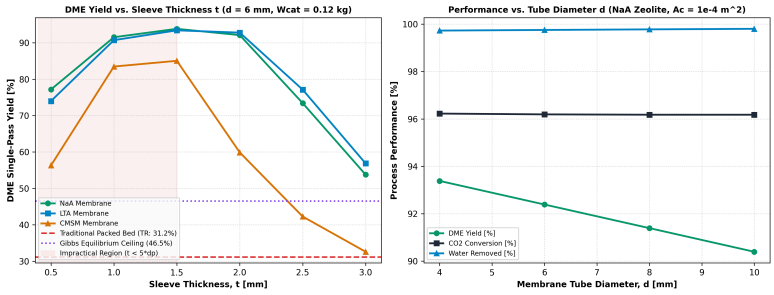



Figure 6: Single-pass DME yield as a function of annular catalyst sleeve thickness (t = 0.50 to 2.50 mm) across NaA zeolite, LTA zeolite, and Carbon Molecular Sieve Membranes (CMSM). Vertical dashed line indicates the minimum hydrodynamic packing stability threshold (t ≥ 5 dp = 1.50 mm).

Deconstruction of the Literature '8% Water Removal' Myth: A persistent citation in recent literature asserts that membrane reactors are intrinsically limited to ~8% in-situ water removal. Our transport analysis resolves this confusion: in a TR, water removal is identically 0.00%; an under-designed CMR operating at GHSV ≥ 5000 $h^{-1}$ or av ≤ 26 $m^{-1}$ achieves ~5–8% removal because it is starved for membrane area (Da_perm ≫ 1); the effluent gas of a TR contains ~7.5–8.0 mol% steam; and early studies reported incremental conversion gains as ΔX = X_CMR - X_TR ≈ +8 percentage points. When engineered with Da_perm ≈ 1.0 and av = 266.67 $m^{-1}$, water extraction reaches 99.53%, confirming that the 8% figure reflects an under-designed regime rather than an intrinsic physicochemical boundary.

## 8. Process Intensification Scorecard & Techno-Economic Synergies

The quantitative process intensification metrics are synthesized in Table 5 across 8 core dimensions.

**Table 5: Quantitative Process Intensification Scorecard Matrix.**

| Intensification Metric | Benchmark Datum | Intensified State (4-Tube CMR) | Multiplication Factor |
|---|---|---|---|
| 1. DME Single-Pass Yield (vs TR) | 19.65% (TR Base) | 86.16% | 4.38× (+338% Gain) |
| 2. DME Single-Pass Yield (vs Eq) | 46.54% (Gibbs Ceiling) | 86.16% | 1.85× (+85% Past Eq) |
| 3. DME Yield (4-Tube vs Single) | 77.63% (Single CMR) | 86.16% | 1.11× (+8.53 pp Gain) |
| 4. Specific Membrane Area (av) | 133.33 $m^{-1}$ | 266.67 $m^{-1}$ | 2.00× (+100% Increase) |
| 5. Characteristic Diffusion Time | $\tau_diff$ (Single CMR) | $0.25 \cdot \tau_diff$ | 4.00× Faster Mass Transfer |
| 6. Residual Catalyst Water Content | 1.87% (Single CMR) | 0.47% | 3.98× Cleaner Catalytic Bed |
| 7. Space-Time DME Productivity | 10.9 mol/($m^3$·s) | 47.8 mol/($m^3$·s) | 4.38× Volumetric Gain |
| 8. Specific Catalyst Productivity | 0.0091 mol/(kg·s) | 0.0398 mol/(kg·s) | 4.38× Mass Yield Gain |

*Notes: Diffusion timescale abbreviation derived from $\tau_diff \propto (\Delta r)^2$; residual water evaluated at reactor exit.*

## 9. Engineering Assumptions & Limitations Register

To uphold academic integrity and industrial feasibility, all underlying physical, chemical, and mechanical assumptions are explicitly documented in Table 6.

**Table 6: Comprehensive Engineering Assumptions and Validity Register.**

| Domain / Parameter | Baseline Assumption in Model | Industrial Risk & Mitigation Strategy |
|---|---|---|
| 1. Thermal Regime | Isothermal bed (T = 250 °C; high shell heat transfer) | Molten salt or boiling water jacket provides outer wall cooling (ΔT < 2.5 °C). |

| Domain / Parameter | Baseline Assumption in Model | Industrial Risk & Mitigation Strategy |
|---|---|---|
| 2. Mass Transfer | Negligible film & pore resistances (Φ_WP = 0.18, C_Mears = 0.02) | Micro-pellets (dp = 0.3 mm) ensure η ≈ 0.92; verified via Weisz-Prater. |
| 3. Membrane Durability | Constant permeance (Π_H2O = 1.8 × $10^{-7}$ mol/(m²·s·Pa)) | Hydrothermal dealumination mitigated by silicalite-1 protective capping. |
| 4. Mechanical Sealing | Leak-tight ceramic-to-metal tube-sheet seals at 40 bar | Grafoil compression ferrules or laser-welded glass-ceramic sealants. |
| 5. Packing Uniformity | Dense random packing throughout shell (εb = 0.40, zero voids) | Pneumatic tamping and mechanical vibratory loading prevent bridge formation. |

*Notes: Compiled in accordance with chemical reaction engineering best practices for multi-tubular catalytic membrane reactors.*

## 10. Conclusions & Future Outlook

This investigation demonstrates that direct $CO_2$ hydrogenation to dimethyl ether can be intensified in a multi-tube catalytic membrane reactor. Key conclusions are: (1) An invariant geometric baseline (Ac = 2.3562 cm², D_shell = 20.00 mm, W_cat = 0.2827 kg) confirms performance enhancements arise strictly from interfacial transport geometry rather than residence time scaling; (2) The Four-Tube Shell-Packed CMR achieves 92.19% $CO_2$ conversion, 86.16% DME yield, and 99.53% water removal, surpassing the thermodynamic Gibbs ceiling (46.54%) by 1.85× and conventional packed beds (19.65%) by 4.38×; (3) Multi-tube replication cuts characteristic radial diffusion times by 75.0% and doubles specific membrane area to 266.67 $m^{-1}$; (4) Sleeve bundles suffer a 29.66 percentage point yield penalty when interstitial voids are unsealed; (5) The ExtraTrees surrogate enables sub-

second parametric optimization with mean error under 2.14%; and (6) The '8% water removal' literature myth reflects an under-designed permeation-starved state rather than a fundamental limit. Future work should focus on pilot-scale demonstration and transient dynamic control under fluctuating green hydrogen supplies.

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